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Ind. Eng. Chem. Res., **Just Accepted Manuscript** • DOI: 10.1021/acs.iecr.7b03417 • Publication Date (Web): 07 Sep 2017

ACS Publications

Industrial & Engineering Chemistry Research is published by the American Chemical Society. 1155 Sixteenth Street N.W., Washington, DC 20036

Improved Oxygen Reduction Reaction Performance of Co Confined in Ordered N-doped Porous Carbon Derived from ZIF-67@PILs

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ABSTRACT

Diffusion and activation of reaction species are two critical factors of oxygen reduction reaction (ORR). Diffusion property is essentially dominated by transmission of reaction species in the channel of catalysts. Herein, we report a facile method to prepare ordered-carbon coated Co nanoparticle (Co@O-NPC). Specifically, we mixed ZIF-67 MOF microcrystal into ionic liquid monomer and resulted in ZIF-67@PILs composites. Subsequently, Co@O-NPC was obtained by the high-temperature pyrolysis of ZIF-67@PILs. Electrochemical measurements show that it possesses superior ORR property comparing to commercial Pt/C. A combination of molecular dynamics (MD) calculations and contrast experimental results reveal that the ordered porous carbon structure can improve the diffusivity of reaction species more effectively than disordered carbon, thereby enhancing ORR property. Moreover, DFT calculation results demonstrate that the catalytically active sites located on the cavity bottom of Co@O-NPC, and uncover synergistic effect between Co nanoparticles and N-doped porous carbon.

KEY WORDS: Oxygen reduction reaction; DFT; MOFs; Diffusion; Co nanoparticle

1. INTRODUCTION

The latest decade has witnessed a surge of research activities regarding high performance proton exchange membrane fuel cell (PEMFC),[1-6] largely stimulated by envisioned technological application. And the efficiency of PEMFC usually relies on the critical oxygen reduction reaction (ORR) at the cathode due to its six-magnitude slower rate than oxidation reaction in anode.[7-13] Motivated by this fact, traditional pure platinum[14-21] and platinum-transition metal alloy[22-29] have been rapidly developed as ORR electrocatalysts. In spite of good catalytic performance, however, it is the kinetically sluggish characteristics of ORR that make high-loading requirement of high-costly platinum, hence hindering the wide commercialization of PEMFC. Consequently, much effort has been made to investigate the substitutes of high-costly Pt-based catalysts both experimentally and theoretically, such as heteratom-doped porous carbons,[9-11, 30, 31] metal oxide, metal carbide and so on.[15, 16, 21, 23, 24, 32]

Within this context, transition metal supported on porous carbon composites (TMPC), due to its low price and promising catalytic property, have captured enormous attentions, e.g. Co/N-doped porous carbon,[33-38] CoN-codoped graphene,[39-42] Fe/N-doped graphene.[43-51] However, although a variety of supported transition metal catalysts have been reported, the accurate structure-function relation of ORR property remains unclear hitherto. This hiatus can be attributed to the diversity of catalytically active sites (such as surface of metal nanoparticle, carbon supports and interface) and complex porosity of TMPC. Essentially, it is well-known that both diffusion and activation of reaction species have a remarkable impact on the performance (specific current density and on-set potential) of ORR. Despite active sites are very critical to catalytic reaction, diffusion of reaction species before and after reaction also has a significant impact on the whole process. In this regard, diffusivity is dominated by porous characteristic as well as adsorption behavior. The effect of enhanced diffusion on catalytic property were investigated in previous experimental and theoretical studies, *e. g.* biomass pyrolysis,[52] methanol oxidation [53] and deposition mechanism of precious metal nanoparticles[54]. However, most studies mainly concentrate on the effect of catalytically active sites in ORR, yet few reports focus on the role of reaction species diffusion.

To explore the effect of diffusion on ORR, we herein prepared ordered-carbon coated Co nanoparticle (Co@O-NPC) and disordered-carbon coated Co nanoparticle (Co@DO-NPC) by using the same ZIF-67 MOF microcrystal as Co source. Electrochemical measurements uncover the better ORR performance of Co@O-NPC than Co@DO-NPC. A combination molecular dynamic calculations and experimental results reveal that the ordered porous structure can effectively improve the diffusivity of O_2 and H_2O molecules and thereby enhance the performance of ORR. Furthermore, molecular-scale DFT calculation results identified the catalytically active sites of Co@O-NPC and confirmed the synergistic effect between Co metal and N-doped porous carbon.

2. EXPERIMENTAI AND COMPUTATIONAL SECTION

2.1. Materials

1-vinyl-3-ethylimidazolium tetrafluoroborate ($(VEIm)BF_4$) was purchased from Lanzhou YuLu Fine Chemical Co., Ltd. Both cobalt nitrate hexahydrate ($Co(NO_3)_2{\cdot}6H_2O$) and 2-Methylimidazole were purchased from Alfa Aesar 2,2´-Azobis (isobutyronitrile) (AIBN) and N,N-Dimethylformamide (DMF) were obtained from TCI Development Co., Ltd. Methanol, ethanol and sulfuric acid were gained from official suppliers and used directly without further purification.

2.2. Synthesis

Synthesis of ZIF-67 microcrystal particles: In a typical synthesis, cobalt nitrate hexahydrate (1.746 g) and 2-methylimidazole (1.970 g) were firstly dissolved separately in a mixed solution of methanol (20 mL) and ethanol (20 mL). Next, the later solution was added largo into the former solution with continuous stirring until the two solutions were mixed completely. The mixed solution was left stewing for 24 h. Finally, the precipitate was acquired by centrifugation and washed with ethanol for several times and dried at 80 ^{o}C for 12 h.

Synthesis of PILs: The procedure of the synthesis of PILs is similar to the progress of the preparation of ZIF-67@PILs described in the next part except remove ZIF-67 from the raw materials.

Synthesis of ZIF-67@PILs: $(VEIm)BF_4$ (2.000 g), ZIF-67 (0.200 g) and DMF (30 ml) were added successively into a pre-prepared 250 mL round-bottomed flask. Then the mixture was first treated with ultrasonication for 1 h to form homogeneous emulsion. Then, AIBN (10 mg) was dissolved in DMF (10 mL), the clear solution was added to the homogeneous emulsion and ultrasonicated for another 10 mins. The final solution was purged with N_2 for at least 30 mins to get rid of the dissolved air and then lowered into a 65 ^{o}C oil bath with continuous stirring for 8 h. Pour the solution into acetone when it is hot and collect the solid product by filtration, dried at 60 ^{o}C overnight.[40, 55]

Synthesis of Co@O-NPC, NC and Co@DO-NPC: To prepare Co@O-NPC, ZIF-67@PILs was calcined in N_2 atmosphere at different temperatures ranged from 500 to 900 ^{o}C for 4 h, Subsequently, the as-annealed powders were treated with 0.5 M H_2SO_4 for 10 h. The final product was then collected by washing with distilled water for several times, and finally dried overnight at 80 ^{o}C, donated as Co@O-NPC-600, Co@O-NPC-700 and Co@O-NPC-800, respectively. For comparison, pure ZIF-67 particles and PILs were used as precursors for synthesizing carbon. The obtained catalysts were donated as Co@DO-NPC-700 and NC, respectively.

2.3. Electrochemical measurements

The electrochemical measurements were conducted with a CHI660D electrochemical workstation (Chenhua Instruments Co., Shanghai) workstation in a three-electrode system at room temperature. The working electrode was rotating disk electrode (RDE), and a Pt wire and an Ag/AgCl (3 M KCl) electrode were used as the counter and reference electrode, respectively. Before each measurement, the KOH solution (0.1 M) was saturated with N_2 (99.999%) or O_2 (99.999%) for at least 30 mins. The electrocatalyst inks were obtained by adding 2 mg of sample and 900 μL ethanol with 100 μL Nafion solution (5% wt) by means of the assistance sonication of at least 30 mins to acquire a homogeneous ink. And then 10 μL of the ink was dropcast onto the glassy carbon disk (diameter 4 mm) of a rotation disk electrode to achieve a catalyst loading of 0.16 mg cm^{-2}. The commercial Pt/C electrode was prepared with the same method for comparison. All potentials were converted to the potentials referring to the reversible hydrogen electrode (RHE), according to $E_{(RHE)} = E_{(Ag/AgCl)} + 0.059 \times pH + E^{\theta}_{(Ag/AgCl)}$ ($E^{\theta}_{(Ag/AgCl)} = 0.1976$ V). Prior to the start of measurement, the electrolyte was bubbled with O_2 for more than 30 minutes. The CV measurements were carried out in O_2- saturated 0.1 M KOH solution with a scan rate of 100 mV s^{-1} from 0.2 to 1.2 V (V *vs*. Ag/AgCl). RDE tests for the ORR were also performed in O_2-saturated 0.1 M KOH solution with the rotation rate ranged from 400 to 2500 rpm and the scan rate was 50 mV s^{-1}. The electron transfer number (n) during the ORR process was calculated from the slopes of linear lines according to the following Koutecky – Levich (K – L) equation:

$$1/J = 1/J_L + 1/J_K = 1/B\omega^{1/2} + 1/J_K$$

$$B = 0.62nFC_0D_0^{2/3}V^{-1/6}$$

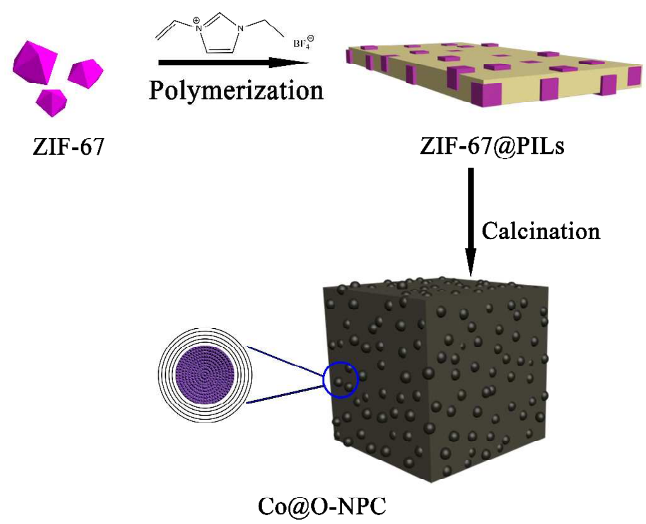
where J is the measured current density on RDE, J_K is the kinetic current density, J_L is the diffusion-limiting current density, B is the reciprocal of the slope, ω is the angular velocity, n is the transferred electron number, F is the Faraday constant (96485 C mol^{-1}), C_0 is the saturated concentration of O_2 in 0.1 M KOH (1.2×10^{-6} mol cm^{-3}), D_0 is the diffusion coefficient of O_2 in 0.1 M KOH (1.9×10^{-5} cm^2 s^{-1}), and V is the kinetic viscosity of 0.1 M KOH (0.01 cm^2 s^{-1}).

2.4. Computational details for DFT and MD

Spin polarized DFT calculations were performed in the VASP[56, 57] software. Exchange and correlation (XC) term in Kohn-Shame equation was approximately treated by Perdew Burke Ernzerhof (PBE)[58] in terms of the gradient of electronic density. Projector-augmented wave (PAW)[59]-based pseudopotential, featuring the greater computational efficiency and high accuracy, was used to the interaction between ions and electrons. Specifically, outer electrons of Co, C, O, N and H were explicitly treated as valence electrons. Plane wave function with kinetic energy less than Ecut of 550 eV is included in basic set. K-pointer in Brillouin zone was set to 4×4×2 grid, being enough accuracy to calculate total energy. The self-consistent field calculations were finished upon the energy difference less than 1.0×10^{-5} eV. The geometrical optimization will be not stop until Hellmann–Feynman force per atom being less than 0.02 eV/Å. Typically, The adsorption energy were typically obtained based on the formula: E_a= E(X-Co-NPC)- E(X) – E(Co-NPC) (X= O_2, O, OH, OOH). In view of practical condition (pH = 13, p = 1 bar, T = 298K), the free-energy difference (ΔG) was calculated according to following formula[60, 61]: $\Delta G= E_a + \Delta ZPE - T\Delta S - neU$. Specifically, n and U represents the electron number in elemental reaction and equilibrium potential, respectively. And zero-point correction energy (ΔZPE) and the reaction-entropy change (ΔS) were derived from the vibrational calculation of and standard table of gas molecules, respectively. Herein, the temperature independence of enthalpy (ΔH (298K) = ΔH (0K)) was approximately used in calculation.

MD simulations were adopted to analyze the O_2 and H_2O diffusion in the confined nanochannel by two graphene or amorphous carbon sheets. The surface areas of the sheet are 42.61 Åx36.90 Å for graphene and 42.73 Åx42.73 Å for amorphous carbon. The nanochannel was then filled with O_2 or H_2O molecules with density of 0.5g/cm^3, and the width of the channel could be adjusted by changing the number of molecules. Nanosheets were fixed during the simulation to prevent any move. The DREIDING force field [62] was applied for describing the interatomic interactions. The philosophy in this force field is to use general force constants and geometry parameters based on simple hybridization considerations rather than individual force constants and geometric parameters that depend on the particular combination of atoms involved in the bond, angle, or torsion terms. Each simulation is carried out for a total time period of 100 ps with a time step of 1 fs. The atomic structures are equilibrated in a NVT ensemble using the Nosé-hoover thermostat at 298 K. All subsequent MD simulations are performed under the same thermodynamic control. Periodic boundary conditions (PBCs) are applied in all directions and MD simulations were carried out using Forcite codes embedded in Materials Studio software.

3. RESULTS AND DISCUSSION



Scheme 1. Synthetic procedure of Co@O-NPC.

3.1. Structural Characterization

As schematically illustrated in scheme 1, both ZIF-67 micro-crystal particles and IL monomers were dispersed and further polymerized into “nougat” composite materials, which poly ionic liquids (PILs) orderly coated ZIF-67 (ZIF-67@PILs) micro-crystal. Subsequently, ZIF-67@PILs composite was treated by high-temperature calcination, resulting in Co@O-NPC. In contrast, disordered N-doped carbon coated on Co nanoparticle (Co@DO-NPC) was prepared by calcination of pure ZIF-67 microcrystal .

As displayed in the inset of Figure 1(a), ZIF-67 features a well-defined three dimensional (3D) block-shape microcrystals. A typical scanning electron microscope (SEM) image clearly shows ZIF-67@PILs presenting a kind of “nougat” structure (see Figure 1a), where PILs act as the “milk” and ZIF-67 particles act as the “peanut”. Powder X-ray diffraction (XRD) and Fourier transform infrared (FT-IR) were carried out to identify the polymerization of ionic liquids and the recombination of the PILs and ZIF-67. Satisfactory, the XRD pattern of as-prepared ZIF-67@PILs composites is a superposition of ZIF-67 and PILs patterns (see Figure S6a), well confirmed the formation of ZIF-67 and PILs composite. Meanwhile, the decrease of IR characteristic peak at around 1650 cm^{-1} (see Figure S6b) associating with C=C bond stretching indicated the effective polymerization of ionic liquids.

SEM images (see Figure 1b and Figure S5a) show both Co@O-NPC-700 and Co@DO-NPC-700 keep the 3D block framework similar to ZIF-67 micro-particles. Detailed structural information of the Co@O-NPC-700 component unit is characterized by high-resolution TEM (HR-TEM) (see Figure 1c – f). Apparently, comparing to Co@DO-NPC-700 (see Figure S5b - f), the Co@O-NPC-700 sample is coated with well-organized nano-sized hollow carbon shell with the diameter ranged from 10 to 30 nm. More interestingly, some of the hollow shell is filled with substance, showed as dark points in the TEM images. The low-magnification TEM image of Co@DO-NPC-700 at the 100 nm scale indicated that Co particles are uniformly distributed on the sample with an average size of about 13 nm (see Figure S7). An independent nano-shell is further observed by HRTEM (see Figure 1e and f). The coated particle features an inter-planar spacing of 2.05 Å, corresponding to the (111) crystal plane (2.046 Å) of Co nanoparticle and the ordered carbon layers exhibits an inter-planar distance of 3.40 Å associating with the crystal plane (002) of graphitic carbon structure. Thereby, it is specifically announced that the as-prepared nano-sized unit is carbon-based "crustal" structure where the shell thickness is several nanometers. Element mapping measurements results (see Figure 1h - j), show uniformly distribution of C, N elements and relatively fewer Co element over the selected area.

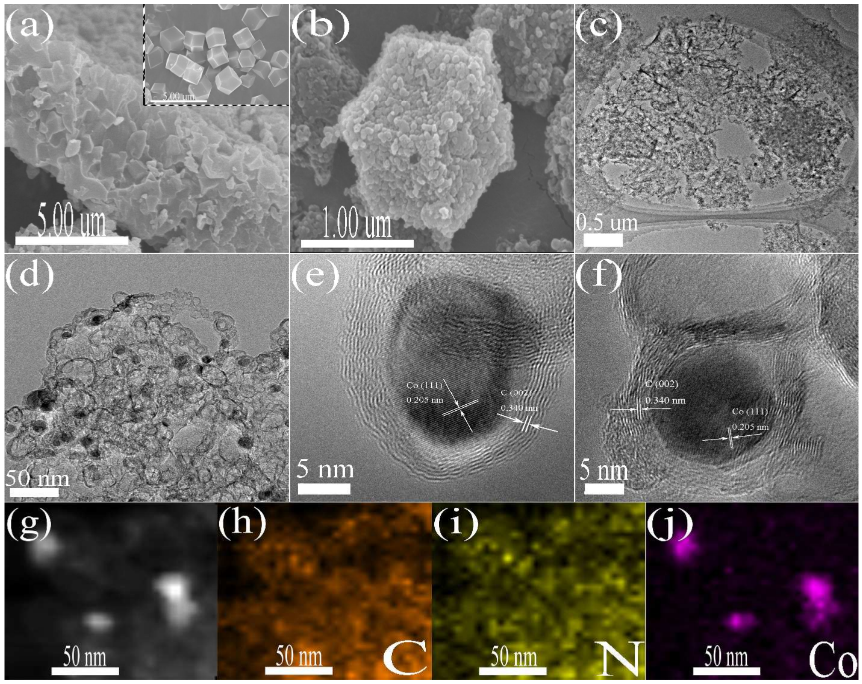


Figure 1 (a) SEM image of the as-synthesized ZIF-67@PILs composites. The inset in (a) shows the SEM image of the as-prepared ZIF-67 particles. (b) SEM image of Co@O-NPC-700. (c) and (d) TEM images of the Co@O-NPC-700 sample. (e) - (f) HRTEM images, and (g) STEM image of the Co@O-NPC-700 sample. (h) – (j) element mapping images of carbon, nitrogen and cobalt, respectively.

X-ray diffraction (XRD) was also carried out to determine the structure properties of the as-prepared Co@O-NPC-700 and Co@DO-NPC-700 samples (see Figure 2a). A diffraction peak at 26.1° corresponding to the (002) plane of graphitic structure was observed in Co@O-NPC-700, in agreement with the ordered carbon layer obtained from the HRTEM results. Meanwhile, the absence of (002) plane in Co@DO-NPC-700 sample finely declared the disorder state of coating carbon was in accord with the TEM result (Figure S5). The other diffraction three peaks at around 44.2°, 51.5° and 75.8° fit well with the diffraction of the metallic cubic-phase Co (JCPDS card no. 15-0806, $a = b = c$ =3.545 Å), affirming the existence of Co NPs presented by TEM images. X-ray photoelectron spectroscopy (XPS) results (Figure 2b - d and Figure S10) determine the chemical states of the elements of Co@O-NPC-700 and Co@DO-NPC-700. The C, N, O and Co elements are all discovered in the two samples, and exhibited an atomic percentage of 87.61%, 6.27%, 5.45%, 0.27% in the Co@O-NPC-700 sample and 80.88%, 9.50%, 4.87% 4.76% in the Co@DO-NPC-700 sample (Table S1), respectively. In addition, inductively coupled plasma optical emission spectrometry (ICP – OES) analysis indicated Co loading of 0.58 wt% on Co@DO-NPC-700 and 0.07 wt% on Co@O-NPC-700. The existence of O is connected with post acid treatment, while both B and F elements were removed from the final product in the process of high temperature calcination and acid treatment. The absence of F and B elements can be attributed to the fragile C-F and C-B bonds under the condition of high-temperature and acid treatment. The high-resolution spectrum of C1s spectrum (see Figure 2b) was further determined by deconvolution and five dominate peaks at around 284.8 eV, 285.8 eV, 286.6 eV, 287.8 eV and 290.2 eV are obtained, corresponding to the carbon state of C-C (sp^2), C-C (sp^3), C=N/C-O, C-O-C/C-N, and -O-C=O peaks, respectively. The N1s spectrum in Figure 2(c) shows the existence of five types of nitrogen species, consistent with pyridinic N at 398.8 eV, pyrrollic N at 399.7 eV, Co-Nx at 399.5 eV graphitic N at 400.8 eV and oxidized N at 402.2 eV, respectively. As shown in Figure 2(d), The Co $2p_{3/2}$ and $2p_{1/2}$ high-resolution spectra is matched with three components consistent with Co(0), Co(II), and the shake-up (satellites) peaks, respectively. The first peak maximum is found at 778.3 eV, corresponding to the Co $2p_{3/2}$ binding energy of Co in a zero valence state. The second peak is revealed at 780.2 eV consistent with the Co^{2+} valence state, in agreement with the binding energy of CoO which is generally presented in the range of 780 – 780.9 eV. The apparent existence of Co^{2+} oxides in the XPS results declares that the Co NPs on the surface of the Co@O-NPC-700 are oxidized in part. It can be explained by the fact that a thin CoO shell could be formed when the Co NPs were exposed in air atmosphere due to the cobalt (0) nanoparticle are sensitive to aerobic environment.[40, 63, 64] The Co-Nx peak was existed at around 782.5 eV and the Co $2p_{1/2}$ states related to the spectrum emerged at around 795 eV. To explore the inner structure of the as-synthesized Co@O-NPC-700 and Co@DO-NPC-700 catalysts, as presented in Figure 2 (e), Raman spectra of Co@O-NPC-700 showed a disordered band at approximately 1342 cm^{-1}, corresponding to the sp^3 defects (D band). Moreover, the crystalline band at around 1590 cm^{-1} was related to the in-plane vibration of sp^2 carbon (G band). Evidently, the *D* band was stronger than the *G* band with an I_D/I_G ratio of more than 1.0 for the Co@O-NPC sample. The N_2 isotherm hysteresis loop of the Co@O-NPC samples reveals a type of IV isotherm, illustrating the mesoporous structure of Co@O-NPC and Co@DO-NPC (Figure 2f and Figure S11b). Meanwhile, the average pore size evaluated by the Barrett-Joyner-Halenda (BJH) method were 4.8 nm for Co@O-NPC-600, 5.3 nm for Co@O-NPC-700, 4.8 nm for Co@DO-NPC-700 and 6.4 nm for Co@O-NPC-800, respectively. According to Brunauer-Emmett-Teller (BET) method, the corresponding specific surface are 205.88 m^2 g^{-1} for Co@O-NPC-600, 219.09 m^2 g^{-1} for Co@O-NPC-700, 263.21 m^2 g^{-1} for Co@DO-NPC-700 and 168.48 m^2 g^{-1} for Co@O-NPC-800, respectively.

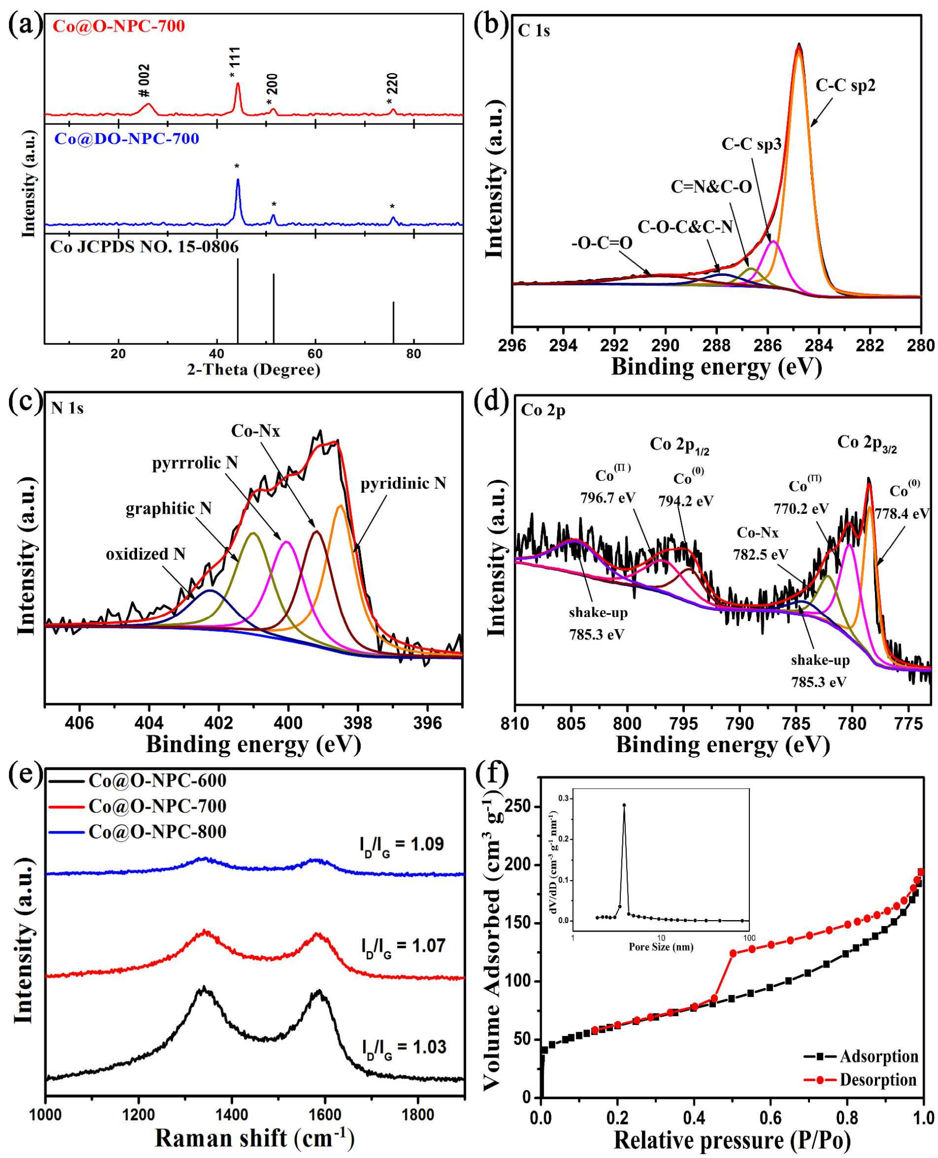


Figure 2(a) XRD patterns of the as-prepared Co@O-NPC-700 and Co@DO-NPC-700 samples. (b) – (d) High-resolution XPS spectra of C 1s (b), N 1s (c) and Co 2p (d) of Co@O-NPC-700. (e) Raman spectra of the as-prepared Co@O-NPC samples with different calcination temperatures. (f) N_2 adsorption –desorption isotherms of the Co@O-NPC-700 sample. The inset in (f) shows the pore size distribution curve.

3.2. ORR Performance

The examination of ORR properties of the as-prepared samples was started with the cycle voltammetry (CV) tests in N_2- and O_2-saturated 0.1 M KOH electrolyte. As displayed in Figure 3(a), compared to the N_2-saturated condition, a remarkable cathodic peak at 0.76 V is detected in the O_2-saturated 0.1 M KOH solution, corresponding to the oxygen reduction process. For comparison, the NC, Co@DO-NPC-700 and the commercial Pt/C catalysts were also measured under the same condition (Figure S14). Among the three samples, the Co@O-NPC-700 shows a more positive ORR peak potential (*ca*. 0.76 V) than NC (*ca*. 0.6 V) and Co@DO-NPC-700 (*ca*. 0.74 V). The electrocatalytic ORR activity of the three samples and the commercial Pt/C were further identified by using liner sweep voltammetry (LSV) curves performed on a rotating disk electrode (RDE) in O_2-saturated 0.1 M KOH solution. As illustrated in Figure 3(c), the Co@O-NPC-700 electrode reveals superior ORR activity with a larger diffusion limited current density (J_d) of 5.3 mA cm^{-2} and a more positive half-wave potential ($E_{1/2}$) of 0.80 V, higher than NC (J_d = 3.6 mA cm^{-2}, $E_{1/2}$ = 0.67 V), Co@DO-NPC-700 (J_d = 4.3 mA cm^{-2}, $E_{1/2}$ = 0.76 V) and commercial Pt/C catalyst (J_d = 5.1 mA cm^{-2}, $E_{1/2}$ = 0.77 V). The obvious increasing of J_d among Co@O-NPC-700 and Co@DO-NPC-700 may be mainly derived from the ordered carbon layer enhanced the diffusion effect in Co@O-NPC-700 sample. Meanwhile, the transferred electron number (n) was confirmed by using the Koutecky-Levich (K-L) equation, and it was found to be 4.0 for Co@O-NPC-700 and Co@DO-NPC-700 in the 0.1 M KOH electrolytes (see Figure 3b and Figure S14d). This announces that both of the Co@O-NPC-700 and Co@DO-NPC-700 electrocatalysts are deferred to the direct 4-electron transfer pathway in the electrocatalytic reduction reaction of oxygen. To investigate the inner ORR reaction process, Tafel plots in the low over-potential range was resulting from the corresponding LSV data (see Figure 3d). By comparing with the NC (114 mV dec^{-1}), Co@DO-NPC-700 (101 mV dec^{-1}) and Pt/C (95 mV dec^{-1}) three catalysts, the Co@O-NPC-700 electrocatalyst presents a smaller Tafel slope of 86 mV dec^{-1} in 0.1 M KOH electrolysts, which indicates a higher transfer coefficient and a faster kinetic process for the ORR process. Furthermore, the EIS spectrum of Co@O-NPC-700 and Co@DO-NPC-700 can be explained on the basis of an equivalent circuit model and the corresponding Nyquist Plots are shown in Figure S15. The charge transfer resistance (Rct) is 24 Ω and 30 Ω, implying larger electrolyte accessibility and higher charge transport capability of Co@O-NPC-700 than Co@DO-NPC-700.

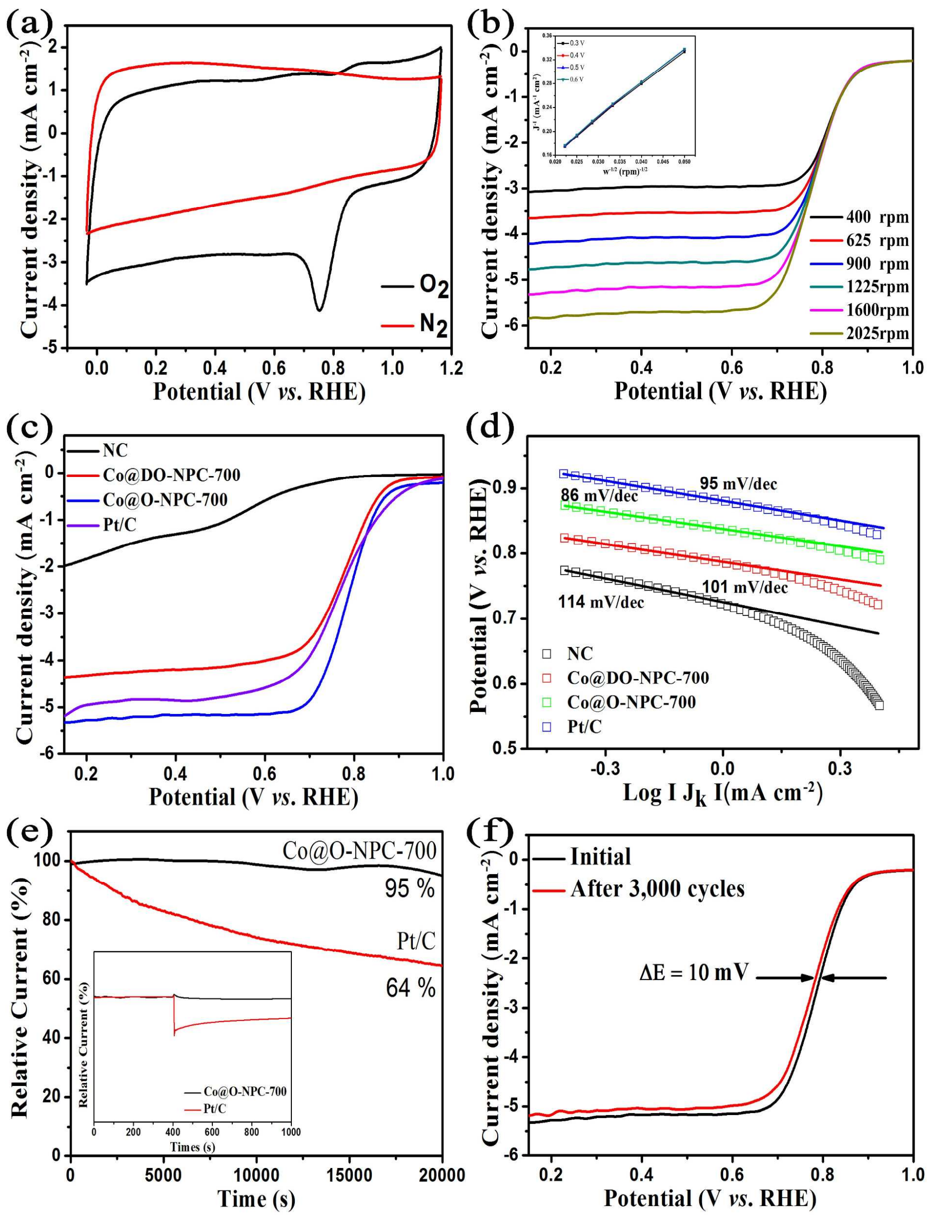
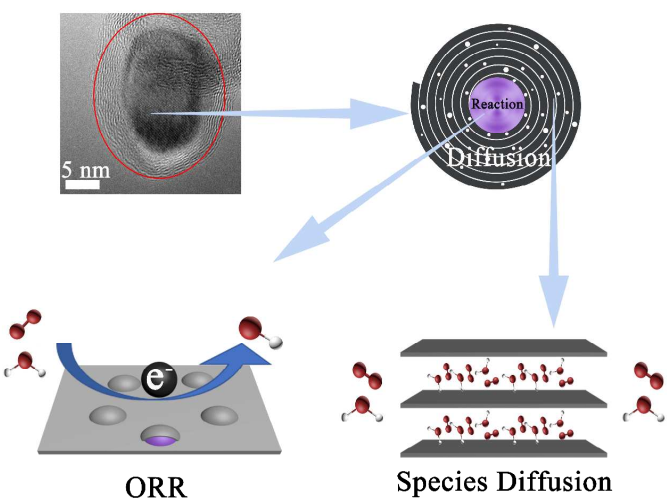


Figure 3 (a) CV profiles (the red curve and the black curve were obtained in N_2-saturated and O_2-saturated 0.1 M KOH solution, respectively.) (b) LSV results different rotation rates. The inset in (b) shows the corresponding K-L plots at different potentials. (c) LSV curves of NC (black curve), Co@DO-NPC-700 (red curve), Co@O-NPC-700 (blue curve) and Pt/C (purple curve). (d) Tafel plots derived from the mass-transport correction of the corresponding RDE data of (c). (e) Chronoamperometric response of Co@O-NPC-700 and Pt/C at 0.6 V (V *vs*. RHE). The inset in (e) shows the tolerance tests to methanol of Co@O-NPC-700 and Pt/C at 0.6 V (V *vs*. RHE). (f) ORR polarization curves (1600 rpm) of Co@O-NPC-700 before and after 3000 cycles.

Furthermore, we evaluated the electrochemical durability of the Co@O-NPC-700 and Pt/C catalysts using the chronoamperometry test at 0.6 V (V *vs*. RHE) in O_2-saturated 0.1 M KOH solution for 20 000 s. As shown in Figure 3 (e), about 95% of the initial current density can be retained for Co@O-NPC-700 at the end of chronoamperometry test, while the loss of current density is about 36% for the commercial Pt/C, suggesting the distinguished stability of the Co@O-NPC-700 catalyst. In addition, accelerated durability test (ADT) was operated between 1.0 and 0.2 V (V *vs*. RHE) at a scan rate of 50 mV s^{-1} for 3000 potential cycles in O_2-saturated 0.1M KOH solution. As illustrated in Figure 3(f), the Co@O-NPC-700 electrode maintains most of the initial activity in the alkaline electrolyte (*ca*. 10 mV negative shifts for its $E_{1/2}$, after 3000 cycles). Furthermore, as shown in the inset of Figure 3(e), methanol resistance test was also conducted for Co@O-NPC-700 to compare with the commercial Pt/C catalyst. Obviously, a marked decrease in the current density is detected of the Pt/C catalyst with the addition of 3 mL of methanol, whereas the current density is almost invariable with the addition of methanol for the Co@O-NPC-700 catalyst. The results manifest that Co@O-NPC-700 has strong immunity toward a methanol crossover effect. Therefore, the prominent improved stability and methanol tolerance of Co@O-NPC-700 catalyst make it an effective candidate for ORR, especially for direct methanol fuel cells.

Essentially, both active sites and species diffusion have an influence on ORR process. Firstly, coordinated Co and pyridine N are mainly active site for Co/N-doped Porous carbon. In this regard, Co@O-NPC-700 and Co@DO-NPC-700 possesses similar active sites because their active components (Co and N) are derived from the same-size ZIF-67 microcrystal precursor. XPS results (see Figure 2 and S10, Table S1) also validate this conclusion. Therefore, the difference of ORR property cannot be attributed to the active sites. In order to investigate whether the process of ORR is dominated by the diffusion effect or the reaction kinetic, CV profiles of Co@O-NPC-700 and Co@DO-NPC-700 samples at different scan rate were conducted in O_2-saturated 0.1 M KOH electrolyte shown in Figure S16 (a – b). As illustrated in Figure S16 (c – d), Peak current is directly proportional to the square root of scanning rate (the fitting results show that the value of R^2 are 0.99971 and 0.99931, respectively). This result validates that the ORR process is dependent on the diffusion effect on the transmission of reaction species in the Co@O-NPC-700 and Co@DO-NPC-700 catalysts. Furthermore, diffusivity is dependent on the pore property and pore structure of transmission channel. For this former, both of the two catalysts have similar specific area (219.09 m^2 g^{-1} for Co@O-NPC-700, 263.21 m^2 g^{-1} for Co@DO-NPC-700) and pore size (5.3 nm for Co@O-NPC-700, 4.8 nm for Co@DO-NPC-700). Therefore, ordered porous structure in Co@O-NPC-700 is more likely to be pivotal factor for the different ORR property.

3.3. Mechanism of Improved ORR property



Scheme 2. Activation and diffusion of Co@O-NPC-700.

Based on aforementioned experimental results, it is concluded that the enhancing ORR performance of Co@O-NPC-700 than Co@DO-NPC-700 is attributed to excellent diffusion of O_2 and H_2O rather than catalytic activation. Therefore, we investigate the catalytic mechanism using DFT calculation, and further identify the effect of diffusion on the ORR property by MD simulation, as shown in scheme 2.

3.3.1 DFT calculation of Catalytic Mechanism

Furthermore, ORR mechanism in Co@NPC (Co@O-NPC-700 and Co@DO-NPC-700) was investigated by mean of spin polarized DFT calculations. In view of experimental results, it is difficult to reproduce the Co nanoparticles of *c.a.* 13.0 nm coating NPC. Therefore, we used the model of 7×7 N-doped graphene with the pore of 6 pyridine nitrogen to describe NPC, and used the Co_{13} cluster matching with this pore to Co nanoparticle. As shown in Figure 4(a), the optimized Co nanoparticle exactly locates in the pore and coordinates with six N atoms. Specifically, two Co atoms chelates with 4 N atoms, while the remaining two Co atoms singly coordinates with two Co atoms, respectively. For Co@NPC structure, the main adsorption sites can be divided into 3 types: the top Co cluster, bottom Co cluster and the interface between Co cluster and NPC. Firstly, we identify the most stable adsorption site of O_2 molecule. Geometrical optimization results of different initial configurations indicate that O_2 cannot adsorb on the interface. And the adsorption energy (−0.65 eV) on the bottom of Co cluster approximated to that (−0.87 eV) on the top of Co cluster. Therefore, it is observed that O_2 probably adsorb on the top and bottom of Co cluster. And the adsorption energy of O, OH and OOH moiety on the bottom of Co cluster (see Figure 4 c - f) are −4.64, −2.29 and −1.68 eV, respectively, while −6.17, −4.33 and −2.84 eV on top of Co cluster, respectively. All reaction species feature chemical adsorption on Co cluster either top or bottom. It can be verified by charge difference of adsorption structure, as shown in Figure S17.

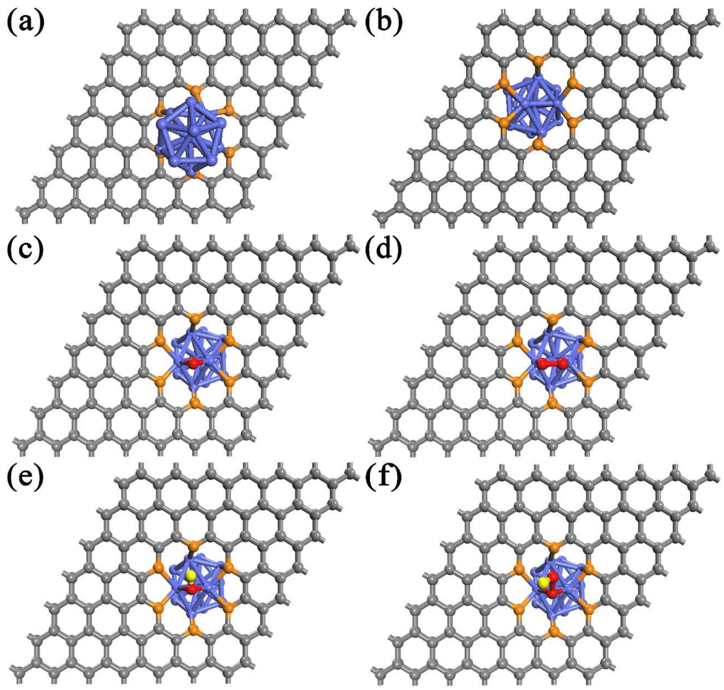


Figure 4 Structure of Co@NPC with top view (a) and bottom view (b) and adsorption structure on the bottom site of O_2 / Co@NPC (c), O / Co@NPC (d), OH / Co@NPC (e) and OOH / Co@NPC (f).

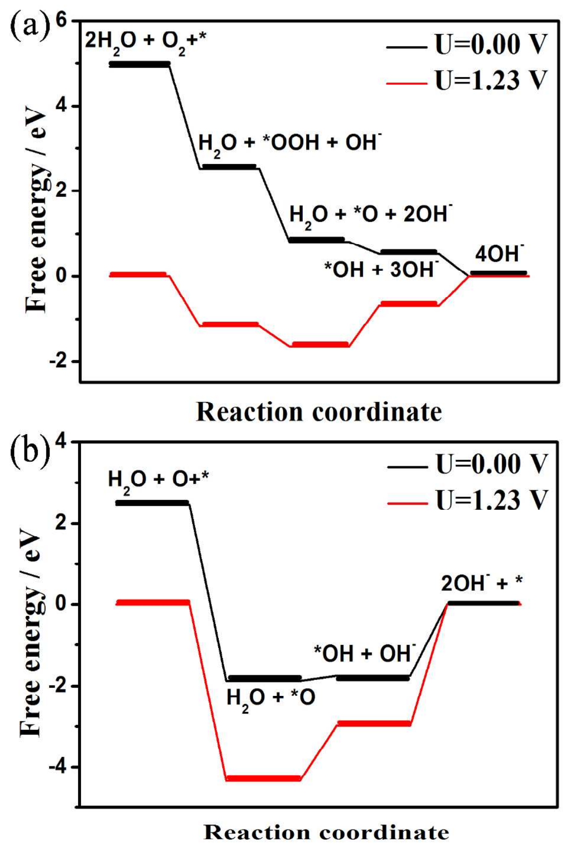


Figure 5 Diagram of free energy at U = 0 and 1.23 V for the bottom site of Co@NPC

under two different mechanisms: associative (a) and dissociative mechanisms (b).

Furthermore, quantitative information of reaction mechanism was probed by calculating Gibbs free-energy diagrams for ORR on Co@NPC. On the cathode in alkaline solution ORR usually shows two types different mechanisms[65], including dissociative and associative mechanisms. Associative mechanism, which can be described as follows:

$$\begin{aligned}
&(1)\ O_2 + H_2O + e^- + * \rightarrow OOH^* + OH^-\\
&(2)\ OOH^* + e^- \rightarrow O^* + OH^-\\
&(3)\ O^* + H_2O + e^- \rightarrow OH^* + OH^-\\
&(4)\ OH^* + e^- \rightarrow OH^- + *
\end{aligned}$$

Dissociative mechanism, in which the steps are the following:

$$\begin{aligned}
&(1)\ \frac{1}{2} O_2 + * \rightarrow O^*\\
&(2)\ O^* + H_2O + e^- \rightarrow OH^* + OH^-\\
&(3)\ OH^* + e^- \rightarrow OH^- + *
\end{aligned}$$

Where * denotes as the free site on the surface.

It mainly consists of five and three elemental steps, respectively, in which the largest difference is concentrated on the formation of OOH* moiety on the catalyst. Herein, we simulated their Gibbs free-energy diagrams at the equilibrium potential of 0.00 and 1.23 V, as shown in Figure 5 and Figure S19. Obviously, at the potential of 0.00 eV, all reaction steps on the bottom site are exothermic, while the formation step of H_2O on the top site is endothermic. Such difference arises from the much larger adsorption energy (−4.33 eV) of OH species than that (−2.29 eV) on bottom site. From the thermodynamical view, therefore, ORR preferably occurs on the bottom site of Co@NPC. Moreover, two types mechanism were identified on bottom of Co sluster. And the rate-determining steps were the formation of H_2O either associative mechanism or dissociative mechanism. The obtained reaction barriers, evaluated by the difference in Gibbs free energy between intermediates, were 0.95 eV for associative mechanism and 3.00 eV for dissociative mechanism, respectively. Therefore, it is concluded that ORR on the bottom site of Co@NPC belongs to associative mechanism of four electrons, consistent with experimentally aforementioned results. Furthermore, Partial DOS of adsorption configuration of reaction species (see in Figure S20) exhibits that 3*d* states of Co, 2*p* states of N and 2*p* states of O either spin-down or spin-up direction effectively overlap in the vicinity of Fermi level. Such phenomenon can demonstrate that odd-electron N can promote spin polarization of Co cluster, shedding light on the indirect effect of N atoms on ORR property[66-69].

Generally, it can be summarized as three points. 1) Active sites of Co@NPC concentrate on bottom site of Co cluster; 2) Catalytic mechanism of ORR is four-electron mechanism; 3) Both pyridine N and Co cluster plays a synergistic role on ORR. Therefore, both Co@O-NPC-700 and Co@DO-NPC-700 feature same ORR activity because of similar electronic structure (elemental population and chemical environment of Co ions).

3.3.2 MD Simulation Regarding Diffusion Property of O_2 and H_2O

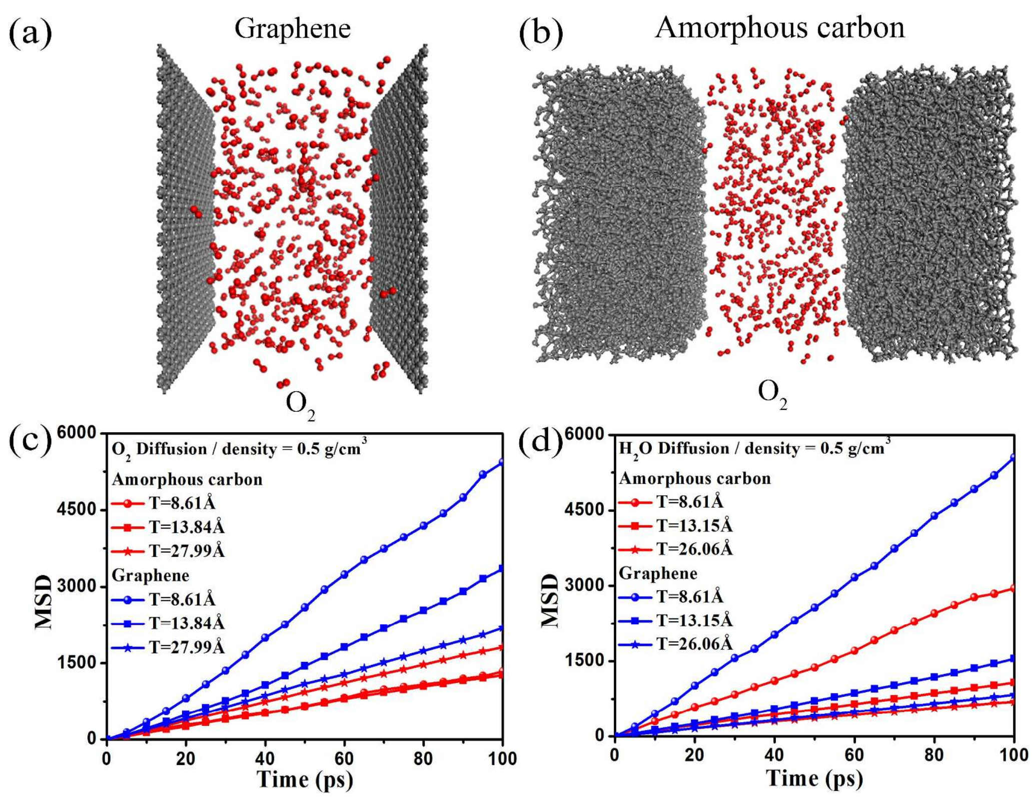


Figure 6 Illustrative snapshots of the system at the end of the 1.1 ns MD simulation for O_2 confined in graphene (a) and amorphous carbon (b). The mean square displacement（MSD）curve of the MD simulation for O_2 (c) and H_2O (d) confined in graphene and amorphous carbon channels.

In order to probe the effect of ordered NPC on the diffusion of reaction species (O_2 and H_2O), molecular dynamics simulation were carried out. It is impossible to experimentally identify the exact position of doped N atoms in porous carbon, so we only consider pure carbon material in this simulation. And the impact factors of diffusion property, *e. g.* nanoslit size and surface structure, were firstly probed by the calculations of mean square displacement (MSD). Specifically, it involves all atomic positions in two kinds of model system with different confined widths. As shown in Figure 6 (a - b), the nanochannel built by pristine graphene can be viewed as an order smooth surface, while the amorphous carbon layers is seen as disorder rough surface. The diffusion coefficient can be typically estimated from the slope of MSD plots via Einstein's relation [70]. The width of confinement is approximately chosen from 8 Å to 28 Å, being close to the size in real experiment. Both O_2 and H_2O filled systems show a significant size dependence of self-diffusion in this width range. Note that these two molecules cannot pass through the channel materials due to the large kinetic diameter comparing with the pore size of graphene or amorphous carbon. With the decrease of width, the diffusion of small molecules in the confined channel is significantly enhanced. Interestingly, the diffusion results also show a strong roughness dependence especially at relatively small width, *e.g.* the MSD for H_2O in the smooth graphene channel is much higher than that in the rough amorphous carbon channel with the width of 8.61 Å in Figure 6 (c) and (d). The size dependence of diffusion can be described by the well-known confinement effect [71] as the maximum width is even less than 3nm. Notably, the width in current system is shorter than the mean free path of molecules under investigation in their gas phase at ambient conditions. The diffusion can then be described as Knudsen diffusion [72] which is a means of diffusion that occurs when the scale length of a system is comparable to or smaller than the mean free path of the particles involved. In the Knudsen regime, the diffusion of small molecules is proved to be strongly dependent on the surface roughness of nanoporous media [73]. And this diffusion in the channel is severely weakened with the increase of roughness factor[74]. This could be adopted to well explain the roughness dependence of molecules diffusion in this work, where the diffusion coefficient in the channel of amorphous carbon is lower than that of graphene channel due to the surface roughness. It verifies that the O_2 or H_2O diffusion in the Knudsen regime can be adjusted by both the confinement size and surface roughness.

Therefore, it is observed that ordered NPC is more favorable for diffusion of reaction species than disorder NPC. Following the results of DFT calculation and MD simulation, we conclude that the differences of ORR activities between Co@O-NPC and Co@DO-NPC are caused by molecules diffusion rather than active sites.

CONCLUSIONS

In summary, we successfully synthesized ordered N-doped porous carbon coating Co nanoparticles (Co@O-NPC) by using the precursors of PILs coated ZIF-67 (ZIF-67@PILs) composites and disordered N-doped porous carbon coating Co nanoparticles (Co@DO-NPC). The resultant Co@O-NPC catalyst possesses excellent catalytic performance for oxygen reduction, which is comparable to commercial Pt/C. A combination of experiment and MD simulation reveals that the more excellent ORR property of Co@O-NPC than Co@DO-NPC arises from better diffusivity induced by ordered NPC transmission channel rather than the difference of active sites. Moreover, DFT calculations unveil the synergistic effect of Co NPs and pyridine N on ORR property. Generally, this study will be helpful to systemically understand the ORR mechanism.

ASSOCIATED CONTENT

Supporting Information

Additional information including structure characterization, electrochemical measurements and theoretical results (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (Grant No: 21671172, 21625604), Zhejiang Provincial Natural Science Foundation of China (Grant No: LY15B010005) and open fund of State Key Laboratory of Structural Chemistry (No: 20170036).

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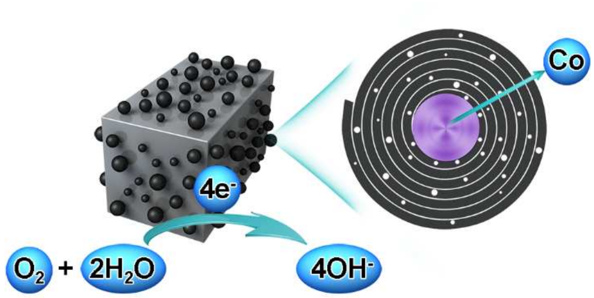

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